

PHYS 487: Lecture # 22

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Lecture Span

1. Decoherence

Let the system be called $|S\rangle$ with some spins. Let $|E\rangle$ be the environment be a collection of $N = \mathcal{O}(10^{23})$ other spins.

Then let the initial state be

$$|S\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Then the total system is

$$|\psi_{SE}\rangle = |S\rangle_i \otimes |E\rangle_i$$

Let the environment be in states $|e\rangle_0$ and $|e\rangle_1$ (both are combinations of spins in E). In principle, the resulting final state S is

$$\text{Tr}_E[\rho_{SE}] = \rho_s = \sum_i \langle e_i | \rho_{SE} | e_i \rangle = \sum_i \langle e_i | \Psi_{SE} \rangle \langle \Psi_{SE} | e_i \rangle \quad (1)$$

This is called a partial trace. We would perform this calculation as follows:

$$\frac{1}{2} \left(\langle e_0 | (|0\rangle |e_0\rangle + |1\rangle |e_1\rangle) (\langle 0| \langle e_0| + \langle 1| \langle e_1|) |e_0\rangle + \langle e_1 | (|0\rangle |e_0\rangle + |1\rangle |e_1\rangle) (\langle 0| \langle e_0| + \langle 1| \langle e_1|) |e_1\rangle \right) \quad (2)$$

This turns out to be

$$\frac{1}{2} [|0\rangle \langle 0| + |1\rangle \langle 1| + \langle e_1 | e_0 \rangle |0\rangle \langle 1| + \langle e_0 | e_1 \rangle |1\rangle \langle 0|] \quad (3)$$

But what makes $\langle e_0 | e_1 \rangle = 0$? If there is at least one spin that is not matching between $|e_0\rangle$ and $|e_1\rangle$. Thus, if $\langle e_1 | e_0 \rangle \approx 0$, then

$$\rho_s = \frac{|0\rangle \langle 0| + |1\rangle \langle 1|}{2} \quad (4)$$

This is the basis of decoherence. That is, the environment can change the density matrix of ρ_s which then changes the initial quantum state. Then, the quantum state is no longer in a superposition of $|0\rangle$ and $|1\rangle$, but rather only in either $|0\rangle$ or $|1\rangle$.

As a starting point,

$$\dot{\rho}_{SE} = -\frac{i}{\hbar} [H_{SE}, \rho_{SE}] \quad (5)$$

Then

$$\rho_{SE}(t + \Delta t) = u_{SE}(\Delta t) [\rho_s(t) \otimes \rho_E(t)] u_{SE}^\dagger(\Delta t) \quad (6)$$

Let there be a "jump operator"

$$L_n = \langle e_n | u | e_0 \rangle \quad (7)$$

Which dictates the probability of the environment jumping from e_0 to e_n . Then

$$\rho_s(t + \Delta t) = \sum_n L_n \rho_s(t) L_n^\dagger \quad (8)$$

We can then expand

$$L_0 \approx \mathbb{I} + \left(-iH_s - \frac{1}{2} \sum_n A_n^\dagger A_n \right) dt \quad (9)$$

Where

$$L_n \approx \sqrt{\gamma_n} dt A_n \quad (10)$$

Here, γ_n is the rate of evolution of E . Then, you can rewrite everything into a differential equation for ρ_s . That is

$$\dot{\rho}_s = -i[H_s, \rho_s] + \sum_n \gamma_n (A_n \rho_s A_n^\dagger) - \frac{1}{2} \{A_n^\dagger A_n, \rho_s\} \quad (11)$$

Measurement – Von Neumann Model

Suppose there is a pointer $|P_i\rangle$. Then we want

$$\alpha |0\rangle + \beta |1\rangle \rightarrow \alpha |0\rangle |p_0\rangle + \beta |1\rangle |p_1\rangle \quad (12)$$

This is also called a "cat state". Since measuring if a particle decayed means that the cat is dead, etc. After applying some time evolution operator of

$$u = \exp(-iH_{int}t/\hbar)$$

Then

$$H_{int} = \hbar g_{int} \sigma_z \otimes p_i \quad (13)$$